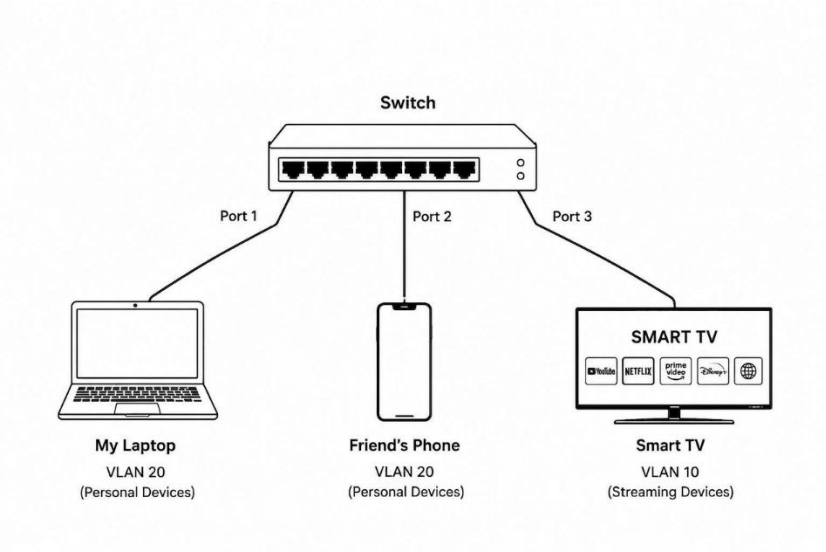
Switching and VLANS:-

Tasks:-

1. Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

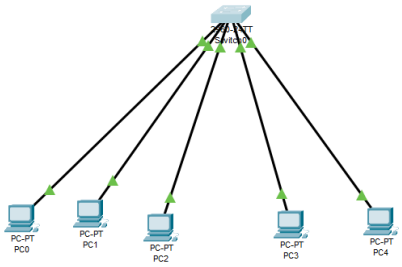
Ans. :-



**2.Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.

Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.**

Ans.:-



Switch>

Switch>ena

Switch#conf t

Switch(config)#vlan 10

Switch (config-vlan)#name vlan10

Switch (config-vlan)#exit

Switch (config)#vlan 20

Switch (config-vlan)#name vlan20

Switch (config-vlan)#exit

Switch (config)#

Switch (config)#interface range fa0/1-3

Switch (config-if-range)#switchport mode access

Switch(config-if-range)#switchport access vlan 10

Switch (config-if-range) #no shut

Switch(config-if-range) #do wr

Switch(config-if-range)#exit

Switch (config)#interface range fa0/4-5

Switch (config-if-range)#switchport mode access

Switch(config-if-range)#switchport access vlan 20

Switch (config-if-range) #no shut

Switch(config-if-range) #do wr

Switch(config-if-range)#exit

Switch (config)#

3. In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

Ans.:-

Switch>

Switch>ena

Switch#conf t

Switch(config)#vlan 10

Switch (config-vlan)#name vlan10

Switch (config-vlan)#exit

Switch (config)#vlan 20

Switch (config-vlan)#name vlan20

Switch (config-vlan)#exit

Switch (config)#

Switch (config)#interface range fa0/1-2

Switch (config-if-range)#switchport mode access

Switch(config-if-range)#switchport access vlan 10

Switch (config-if-range) #no shut

Switch(config-if-range) #do wr

Switch(config-if-range)#exit

Switch (config)#interface range fa0/3-5

Switch (config-if-range)#switchport mode access

Switch(config-if-range)#switchport access vlan 20

Switch (config-if-range) #no shut

Switch(config-if-range) #do wr

Switch(config-if-range)#exit

Switch (config)#

PC3 was successfully moved from VLAN 10 to VLAN 20. After changing its IP address to the VLAN 20 subnet, it can communicate with PCs in VLAN 20 but cannot communicate with PCs in VLAN 10 because there is no inter-VLAN routing.

**4. Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).

Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.**

Ans.:-

Switch>

Switch>ena

Switch#conf t

Switch (config)#vlan 30

Switch (config-vlan)#name vlan30

Switch (config-vlan)#exit

Switch (config)#

Switch (config)#interface range fa0/1-10

Switch (config-if-range)#switchport mode access

Switch(config-if-range)#switchport access vlan 30

Switch (config-if-range) #no shut

Switch(config-if-range) #do wr

Switch(config-if-range)#exit